unfound

PREPARED FOR TIDE · OCTOBER 2026

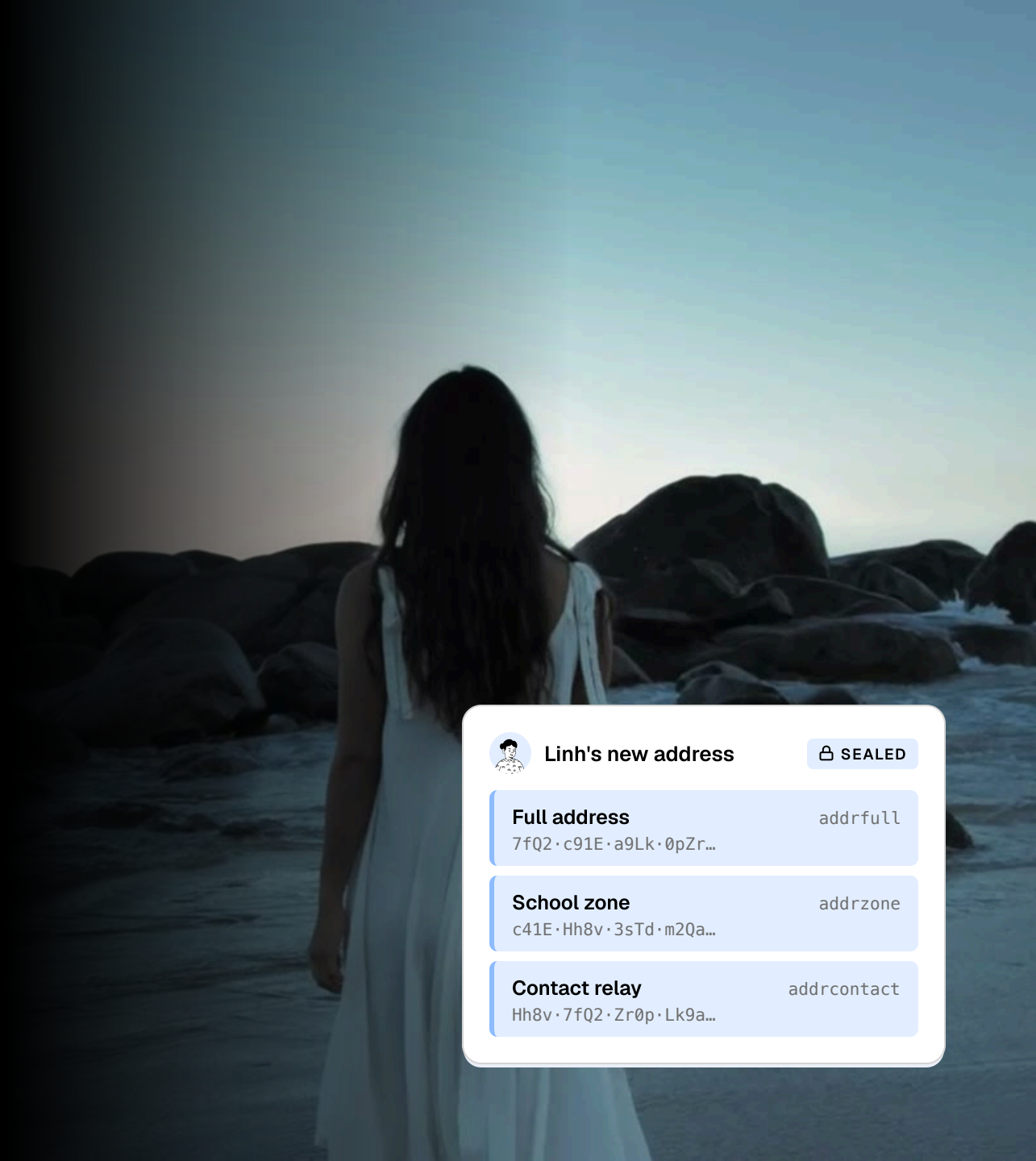
# *Be reached,* not found.

## Every school. Every clinic. No one looks you up.

A safe-address layer for family-violence survivors. Services can reach them after they move. No employee, admin or operator can look them up.



Seven people. One address.
**No master key.**



# Leaving is when the risk peaks. The new address is what the abuser looks for.

AU · PREVIOUS PARTNER

**1.5M**

Australian women have experienced violence from a **previous** partner, nine times more than from a current one. [1]

AU · MOVED OUT

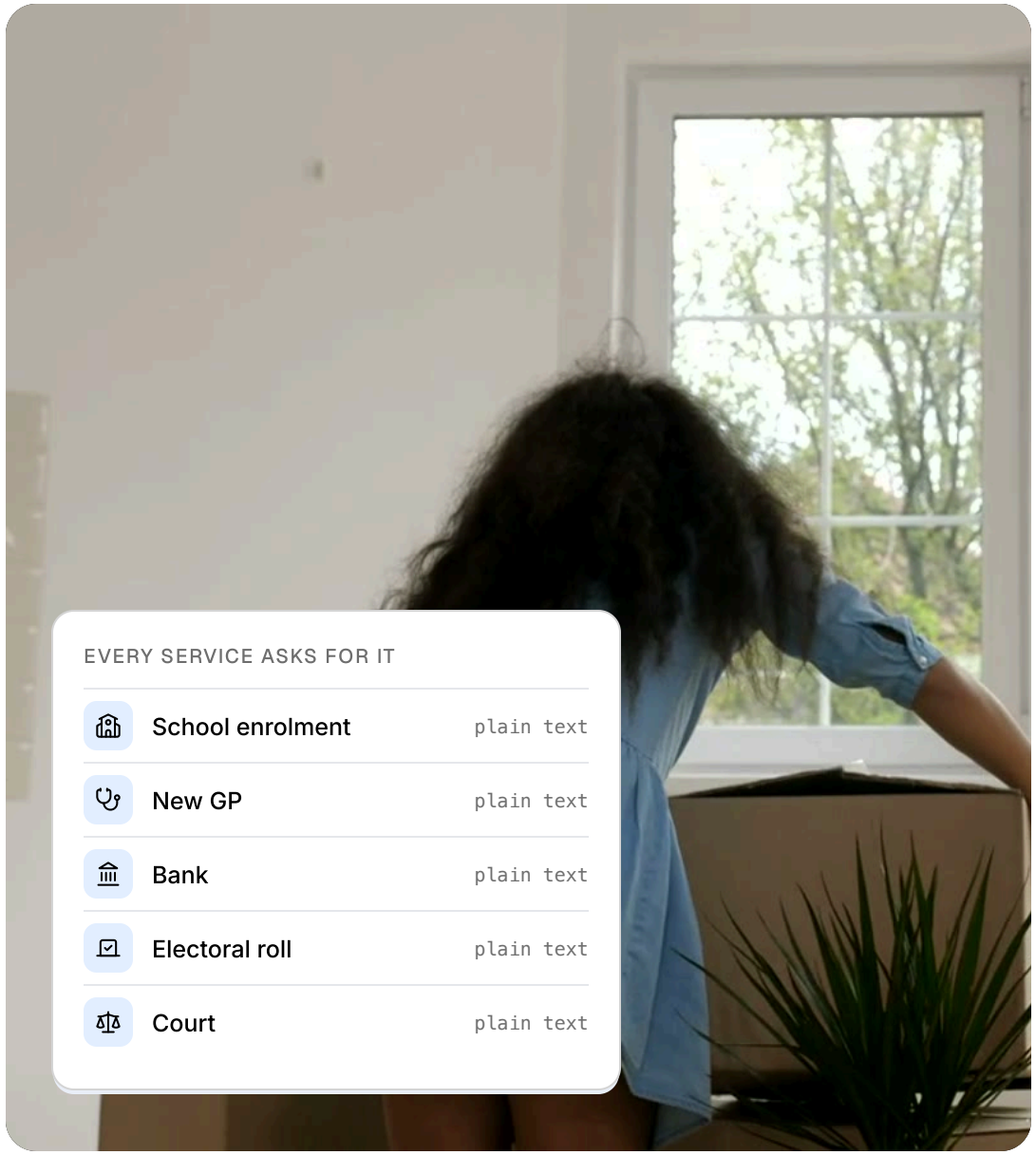
**866,700**

of them moved out when it ended. Each move is a new address in school, GP, bank and court records. [1]

NSW · SEPARATION

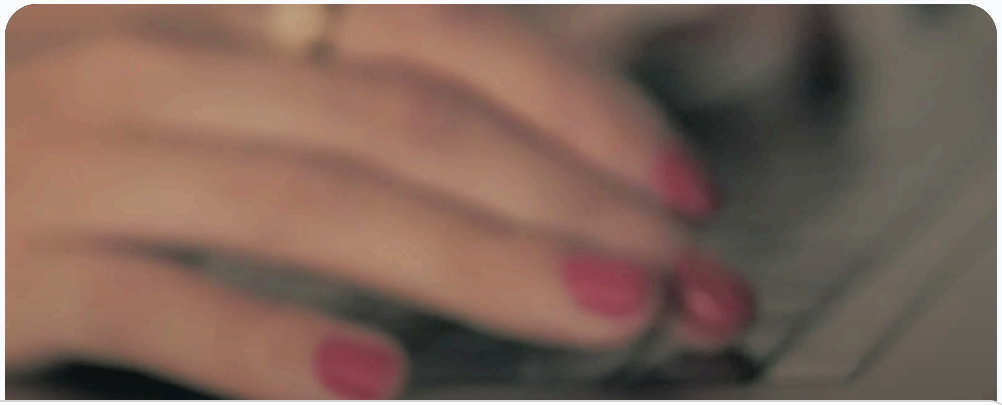
**2 in 3**

women killed by an intimate partner were killed after leaving or while separating. [2]




[1] ABS, Personal Safety Survey 2021–22 (accessed 22 Oct 2026). [2] NSW DV Death Review Team 2021–23, via Canberra Times (accessed 26 Oct 2026). Photo: Mixkit.

# Not a hacker. One person with legitimate access, doing one lookup.



QLD · 7 AGENCIES · 4 YEARS

## 2,215

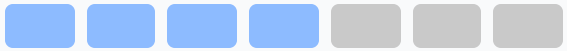
misuse-of-information allegations. Outside police: **4.8×**. [3]




UK REGULATOR · 2022–23

## 4 of 7

organisations reprimanded over victims' data had **sent a safe address to the abuser**. [4]




**Records · admin view**

Dan K. · IT admin · search: "Linh"

LEGACY SYSTEM

| | | |
|---|---|---|
| Nguyen, T. | 14 Albert St, Footscray VIC 3011 | |
| Linh P. | Unit 4, 18 Moreland St, Footscray VIC 3011 | VIEWED |
| Okafor, J. | 2/91 Barkly St, Footscray VIC 3011 | |
| Smith, R. | 7 Hyde St, Yarraville VIC 3013 | |

No approval. No alert. The survivor never knows.

2014 · QUEENSLAND

An officer searched the police database and gave a survivor's new address to her abuser. [5]

2023 · ENGLAND

A council posted a letter with a survivor's new address to her old home. Her ex-partner opened it. [6]

[3] CCC Qld, Operation Impala Ex. 95. [4] UK ICO, 29 Sep 2023. [5] QCAT via Sydney Criminal Lawyers. [6] ICO reprimand, Charnwood BC, 29 Nov 2023. Accessed 22–27 Oct 2026. Records are synthetic.

# Every fix today still trusts someone who holds the key.

STILL TRUSTS A PERSON

### Role-based access + audit logs

Admins configure and override them. Logs show harm after it has happened.

Still readable by: every admin

STILL TRUSTS A PERSON

### Encryption at rest

The provider holds the key, so the operator, and anyone who breaches them, reads everything.

Still readable by: the operator

STILL TRUSTS A PERSON

### Silent elector (AU)

Hides the address on the electoral roll only. Schools, clinics and courts still store it.

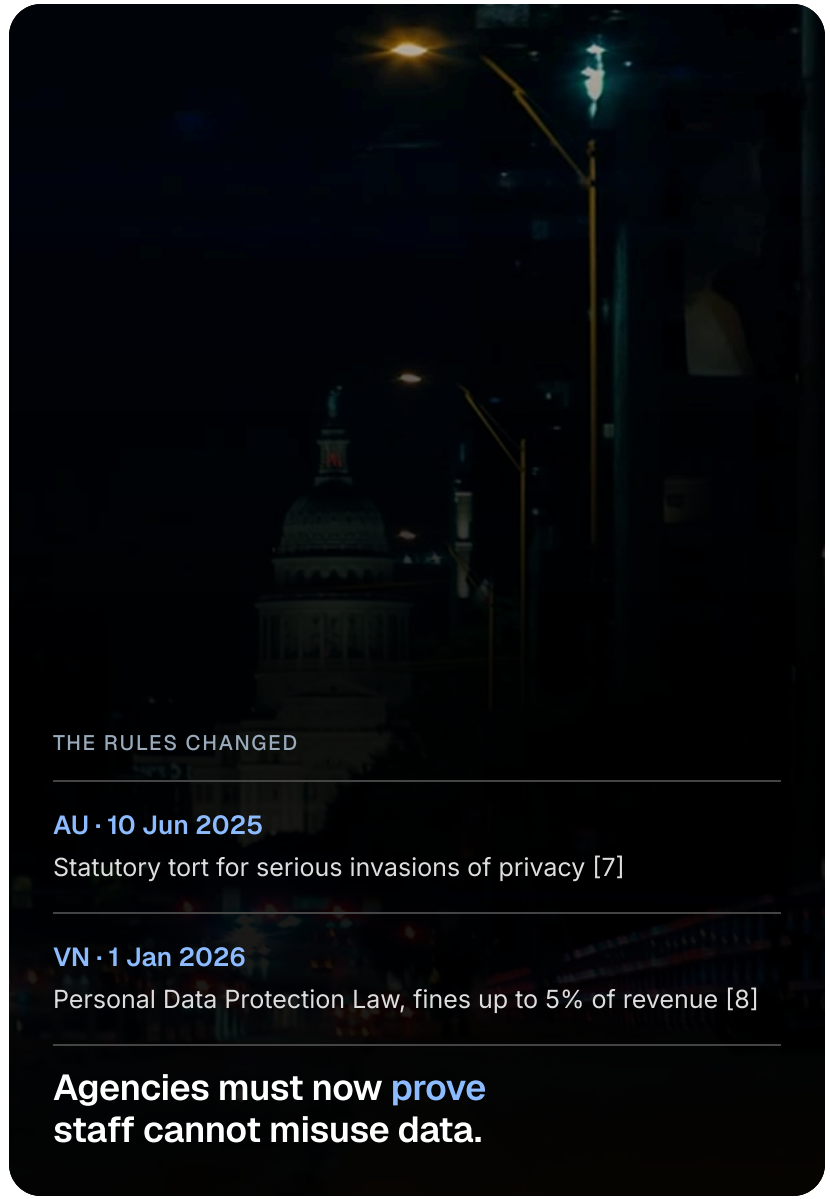
Still readable by: every other service

STILL TRUSTS A PERSON

### Address Confidentiality Programs (US, ~39)

Manual mail forwarding. No cryptographic control, and nothing for the survivor to see.

Still readable by: program staff



THE RULES CHANGED

**AU · 10 Jun 2025**
Statutory tort for serious invasions of privacy [7]

**VN · 1 Jan 2026**
Personal Data Protection Law, fines up to 5% of revenue [8]

**Agencies must now prove staff cannot misuse data.**

[7] Lander & Rogers. [8] Tilleke & Gibbins; Conventus Law. ACP count: NACAP to NASS (2022). Accessed 26 Oct 2026. Photo: Mixkit.

# One address. Three sealed parts.
# Each service opens only its own.



**Westgate Primary** ZONE ✓
In Footscray North zone



**GP clinic** RELAY
Suburb and a relay number



**Court mail** AT PRINT
Street opens as the label prints



**Linh · survivor** SEALED
Sealed once, on her phone

| | |
|---|---|
| **Full address** | addrfull |
| quorum only · one view | |
| **School zone** | addrzone |
| catchment only | |
| **Contact relay** | addrcontact |
| suburb + relay number | |

**Who looked:** every attempt, approval and view



**Const. Reyes** 2 OF 3
One view, then revoked



**Dan · IT admin** 412
Ciphertext; self-grant pending



**Unfound · operator** NO KEY
Stores ciphertext only

SCOPE

# What Unfound protects, and what it doesn't.

Unfound closes one path: services that hold a survivor's address and leak it through a lookup, a permission change or a letter. It is one part of a safety plan, not a replacement for it.

| | |
|---|---|
| VERIFIED ON LIVE TIDE | Licensed realm with QEA. Enrolment blocked by ORK availability, so no live decrypt yet. |
| IMPLEMENTED IN CODE | Sign-in with DPoP, doEncrypt / doDecrypt per tag, AdminAPI change sets, six roles. |
| SIMULATED IN THE PROTOTYPE | The eight-scene story on a mock fabric; relay, court printing and one-view revoke. |

## COVERED 4

- ✓ **Lookups by staff or a compromised account**
  Each opens only its tag; every attempt is logged.
- ✓ **An admin granting themselves access**
  A change set that needs 2 of 3 approvers.
- ✓ **A breach of Unfound's servers**
  Only ciphertext to take, no key beside it.
- ✓ **Letters and forms made through Unfound**
  The street is filled in only at print.

## NOT COVERED 5

- ✕ **Following, GPS trackers, shared location**
  Needs safety planning and tech-safety support.
- ✕ **People who already know the address**
  Friends, family, neighbours.
- ✕ **Copies at services not on Unfound**
  Their plain-text copy keeps the old exposure.
- ✕ **Notes taken after a legitimate view**
  An officer can still write it down.
- ✕ **The survivor's own phone, if taken**
  Reduced by design, not prevented.

# The promise only holds if no one can hold the key, or grant themselves access.



### Survivor's browser

HERMETIC E2EE

```
tc.doEncrypt([
 { data: street, tags: ["addrfull"] },
 { data: zone, tags: ["addrzone"] },
 { data: relay, tags: ["addrcontact"] }
])
```

Sealed before it leaves the phone. Sign-in with BYOiD; tokens bound by **DPoP**.

### Tide Fabric

INDEPENDENT ORK NODES

14 OF 20

Keys exist only as fragments. Before helping decrypt, the ORKs check the caller's tag role: `_tide_addrzone.selfdecrypt` . The network refuses, not our UI.

### Unfound store

CIPHERTEXT ONLY

```
AAFb7fQ2c91Ea9Lk0pZrHh8v3sTdm2QaZr0pLk9aXc4E…
AAGc41EHh8v3sTdm2Qa7fQ2c91Ea9Lk0pZrB1x…
AAHh8v7fQ2Zr0pLk9a3sTdc41Em2QaXw8…
```

No key on our servers to steal, leak or subpoena.

QUORUM-ENFORCED GOVERNANCE

## No one alone.

Every access change is a change set signed by a threshold of admins.



Mai + Ruth approve police break-glass — 2 OF 3 · SIGNED



Dan grants himself the street — 412 QUORUM NOT MET

TIDE IN THE CODE

- TideCloak BYOiD
- DPoP ES256
- doEncrypt / doDecrypt
- 6 tag roles
- AdminAPI change sets
- tide_doctor PASS



*"Emergent Authority, made human."*

One survivor. Seven parties. No one in charge of the key.

06 WHY THIS MATTERS TO TIDE

# Every Tide primitive, in one story anyone understands in a minute.

### A reference use case

Shows what "no one holds the key" means for a real person, not a server.

FABRIC · E2EE · BYOID

### A public-sector door

Schools, health, courts and police: buyers who must now prove insiders cannot misuse data.

GOVERNMENT · HEALTH · JUSTICE

### A governance showcase

Break-glass with witnesses is QEA's clearest demo: two of three sign, one view, revoked.

QEA · THRESHOLD SIGNATURES

### Real product feedback

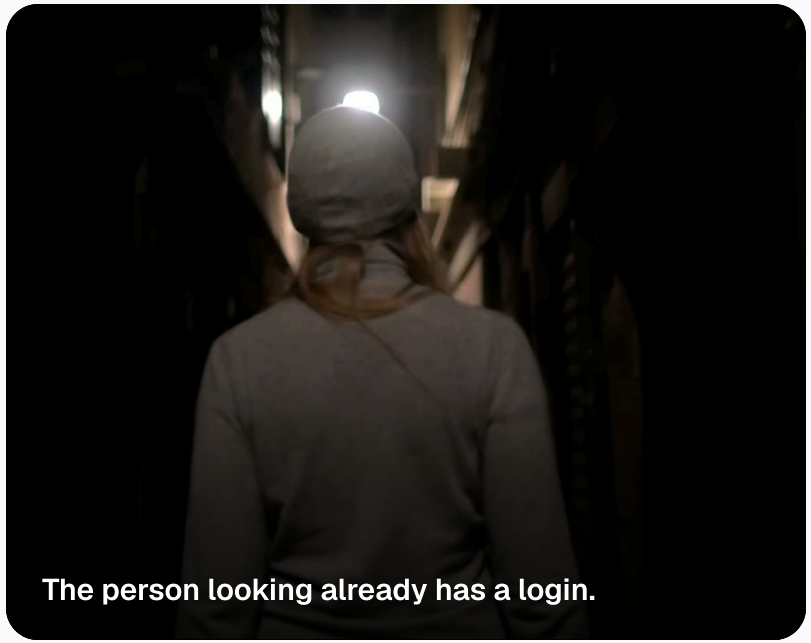
Cross-user decryption, one-time views, cross-org quorums, enrolment health: found by building.

FORSETI · ENROLMENT · SDK

07 THREAT MODEL

# What each adversary actually gets.

Unfound is designed for the people who already have access: staff, admins, and us. Each row is enforced by Tide, not by a policy document.

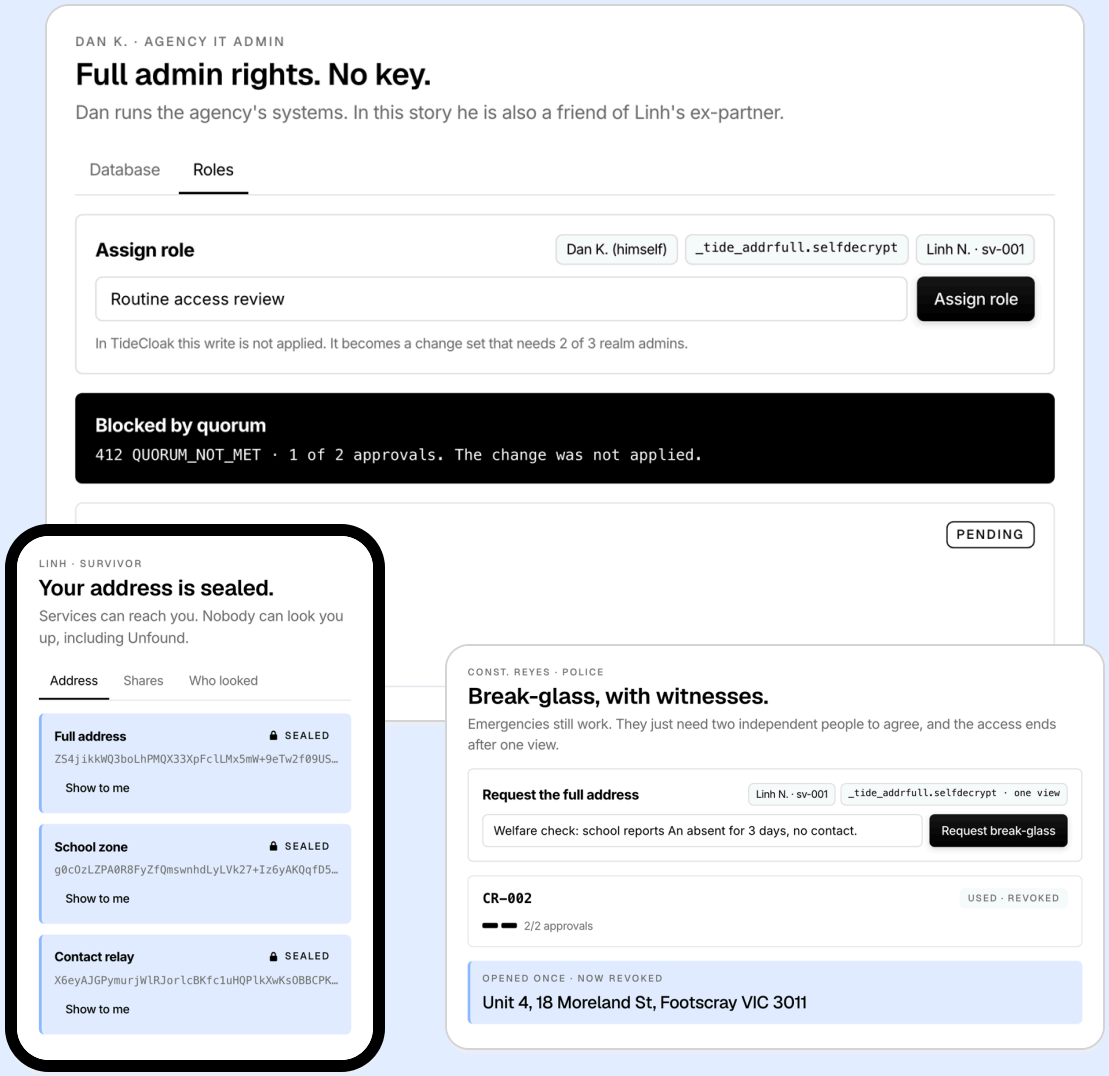



| WHO TRIES | WHAT THEY GET | |
|---|---|---|
| **A curious or bribed staff member** | Only the part their role allows; the attempt lands in the survivor's log | DENIED |
| **A realm admin granting themselves access** | A pending change set, then 412 QUORUM_NOT_MET | PENDING |
| **Whoever runs Unfound's servers** | Ciphertext. No key to steal or subpoena | CIPHERTEXT |
| **A compromised TideCloak server or one ORK** | No usable key: keys exist only as fragments across nodes | FRAGMENT |
| **Someone holding a stolen token** | Nothing from another device: the token is DPoP-bound | REJECTED |
| **Coercion of the survivor's own device** | Out of scope for crypto. Nothing shown by default; quick exit and caseworker recovery on the roadmap | MITIGATED |

# A working prototype, end to end.

| | | |
|---|---|---|
| **7** personas on iPhone, MacBook and iPad | **68s** auto-played story, 8 scenes | **15** Tide operations per run |
| **0** plaintext addresses stored | **6** `_tide_addr*` roles over 3 tags | **2/3** quorum for any access |
| **3** ✓ tide_doctor checks PASS | **1:55** film, rendered from the app | **0** console errors end to end |

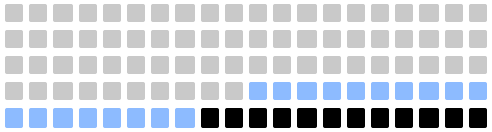
Live path wired: @tidecloak/react 0.14.33, DPoP, doEncrypt/doDecrypt, AdminAPI change sets. First enrolment failed on 22 Oct 2026 when Tide's ORK network answered 3–7 of 20; the public demo runs on a mock fabric with the same rules.



# Tide supplies the hard part. The rest is ordinary engineering, and it is funded.

MARKET · AU SCHOOLS

## 9,653

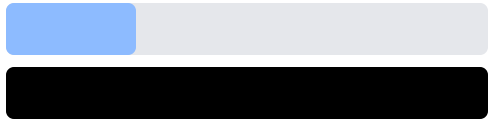


70% government · 18% Catholic · 12% independent

At a hypothetical **A$49 per school per month**: about **A$5.7M a year** from one segment. [9]

FUNDING · AU

## A$925M



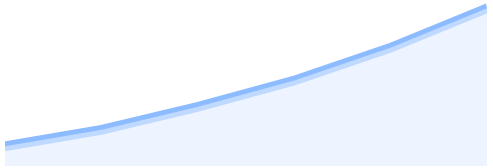
Leaving Violence · A$925M

National Plan · A$3.4B+

Five-year program for people leaving violence, a pathway for funded delivery partners. [10]

PROVEN DEMAND · US

## 6,212

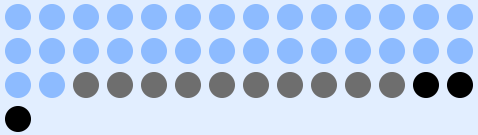


~12% annual growth · over half are school-aged children

Active participants in California's address program alone, run by hand. [11]

PILOT · TIDE FREE TIER

## 100

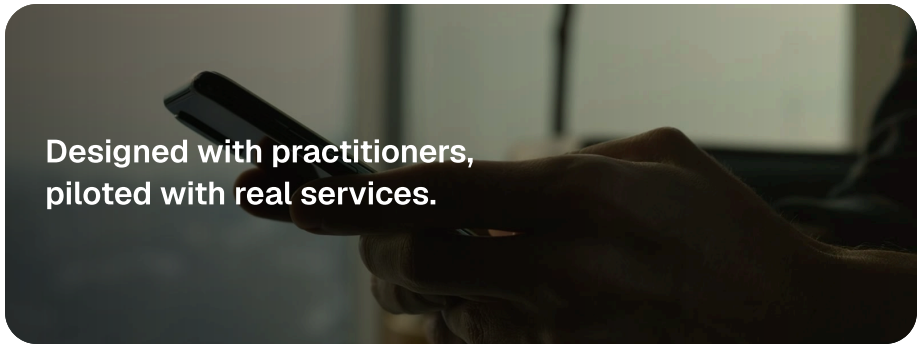


30 survivors · 10 staff · 3 approvers = 43

TideCloak's free tier covers the whole first pilot. [12]

[9] ACARA / ABS Schools 2024; price is a hypothesis to test. [10] DSS Ministers, 2024–25 Budget. [11] CA Secretary of State to NASS, 2023. [12] tide.org/pricing. Accessed 26 Oct 2026.

# From prototype to a supervised pilot in 2027.




**Nov–Dec 2026** · PHASE 0 — **Jan–Mar 2027** · PHASE 1 — **Apr–Sep 2027** · PHASE 2 — **2028** · PHASES 3–4

## Live on Tide

Hosted realm in Australia. Cross-user decryption and one-time views confirmed with Tide.

EXIT CRITERIA

Full story on live Tide; tide_doctor all PASS

## Design partners

One family-violence service, one school, one GP clinic. Practitioner safety review.

EXIT CRITERIA

Signed pilot, PIA, pen test passed

## Supervised pilot

20–30 survivors, 3 services, 3 independent approvers.

EXIT CRITERIA

0 plaintext · 100% of attempts visible · onboarding < 15 min

## Integrate and scale

School and GP connectors, court print relay, Forseti one-time access. Multi-state AU, Viet Nam pilot.

EXIT CRITERIA

Funded contracts in two states

# Help us take Unfound from prototype to pilot.

01 **A hosted realm**
TideCloak in an Australian region, plus the Launch Program.

02 **Enrolment clarity**
What caused the ORK failures on 22 Oct, and which health signals to watch.

03 **Two introductions**
A family-violence service and a public agency open to a pilot.

04 **A reference build**
Unfound as a Tide case study once the pilot is live.

FOUR TECHNICAL QUESTIONS

**Q1** Can data sealed by one user on tag X be opened by another user who holds `_tide_X.selfdecrypt`?

**Q2** Can a Forseti contract enforce “decrypt once, then expire” for break-glass?

**Q3** Can one realm's quorum include admins from different organisations?

**Q4** Can change-set history be exported per record for a survivor-facing log?

# *Be reached,* not found.

Survivors stop choosing between help and hiding. Agencies can prove that no single person, including their own admins, can reveal where someone lives.

**Demo, film, code and brief**
github.com/BennedictQuanTon/Unfound-Be-reached-Not-found

built with **TideCloak**

Footage stills: Mixkit (free licence). People: Notionists by Zoish (CC0) via DiceBear. Icons: Lucide (ISC). All records are synthetic.